Definition 1.8

A system of linear equations having at least one solution is said to be **consistent**. A system of linear equations having no solution is said to be **inconsistent**.

Remark

If the number of the equations of a system of linear equations is equal to the number of unknowns of the system, then the coefficient matrix A of the system is a square matrix. Further, if A is a non-singular matrix, then the solution of system of equations can be found by any one of the following methods : (i) matrix inversion method, (ii) Cramer's rule, (iii) Gaussian elimination method.

1.4.3 (i) Matrix Inversion Method

This method can be applied only when the coefficient matrix is a square matrix and non-singular.

Consider the matrix equation

$$AX = B, \qquad \ldots (1)$$

where A is a square matrix and non-singular. Since A is non-singular, A^{-1} exists and $A^{-1}A = AA^{-1} = I$.

Pre-multiplying both sides of (1) by A^{-1}, we get $A^{-1}(AX) = A^{-1}B$. That is, $(A^{-1}A)X = A^{-1}B$.

Hence, we get $X = A^{-1}B$.

Example 1.22

Solve the following system of linear equations, using matrix inversion method:

$5x + 2y = 3,\ 3x + 2y = 5$.

Solution

The matrix form of the system is $AX = B$, where $A = \begin{bmatrix} 5 & 2 \\ 3 & 2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix}, B = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$.

We find $|A| = \begin{vmatrix} 5 & 2 \\ 3 & 2 \end{vmatrix} = 10 - 6 = 4 \neq 0$. So, A^{-1} exists and $A^{-1} = \frac{1}{4}\begin{bmatrix} 2 & -2 \\ -3 & 5 \end{bmatrix}$.

Then, applying the formula $X = A^{-1}B$, we get

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 2 & -2 \\ -3 & 5 \end{bmatrix}\begin{bmatrix} 3 \\ 5 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 6-10 \\ -9+25 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} -4 \\ 16 \end{bmatrix} = \begin{bmatrix} \frac{-4}{4} \\ \frac{16}{4} \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}.$$

So the solution is $(x = -1, y = 4)$.

Example 1.23

Solve the following system of equations, using matrix inversion method:

$2x_1 + 3x_2 + 3x_3 = 5,\quad x_1 - 2x_2 + x_3 = -4,\quad 3x_1 - x_2 - 2x_3 = 3.$

Solution

The matrix form of the system is $AX = B$, where

$$A = \begin{bmatrix} 2 & 3 & 3 \\ 1 & -2 & 1 \\ 3 & -1 & -2 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, B = \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix}.$$

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